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RESEARCH ARTICLE

Co-Simulation Framework for Network Attack Generation and Monitoring

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The research described in this paper is part of the Resilience Through Data-Driven, Intelligently Designed Control (RD2C) Initiative at Pacific Northwest National Laboratory (PNNL). It was conducted under the Laboratory Directed Research and Development Program at PNNL, a multiprogram national laboratory operated by Battelle for the Department of Energy.

ABSTRACT Resilience assessment is crucial for maintaining high availability, security, and quality of service in power grids. However, most current grid research lacks hardware testbed capabilities. Moreover, the integration of distributed energy resources expands the grid's attack surface, necessitating reliable and realistic modeling techniques to be accessible to the broader research community. Consequently, simulation testbeds have emerged to model real-world power grid topologies and evaluate the impact of various disruptions. Existing co-simulation platforms for power grids focus on limited components, such as focusing only on the dynamics of the physical layer. Additionally, many platforms require specialized hardware that may be too expensive for most researchers. Furthermore, not many platforms support realistic modeling of the communication layer, which demands the use of Supervisory Control and Data Acquisition (SCADA) communication protocols like DNP3 for cybersecurity scenario modeling. We introduce Network Attack Testbed in [Power] Grid (NATI[P]G) (pronounced "*natig*"), a stand-alone, containerized, and reusable environment that enables cyber analysts and researchers to execute various cybersecurity and performance scenarios on power grids. NATIG integrates GridLAB-D, a grid simulator, HELICS, a co-simulation framework, and NS3, a network simulator, to create an end-to-end simulation environment for the power grid. We demonstrate use cases by generating a library of datasets for several scenarios. These datasets can be utilized to detect cyberattacks at the cyber layer and develop countermeasures against these adverse scenarios.

INDEX TERMS Cyber-physical systems (CPS), communication networks, telecommunication network, power grid, microgrids.

I. INTRODUCTION

Cyber-physical systems (CPS), such as microgrids, are key infrastructure components that impact social, financial, and national security on a daily basis. Thus, with cyberattacks increasing in sophistication by the day, there is a need to understand various failure scenarios and plan for mitigation strategies for a reliable and resilient power grid operation [1], [2]. Simulation environments have been extensively used to model CPS components, topologies, adversaries, attack sequences, and their impact on the systems [3], [4]. CPSs are complex and interdependent systems, with both discrete and continuous measurements. However, existing attack detection models, such as data-driven intrusion detection and prevention systems, require a high amount of data to train models before they are deployed on the grid. This stymies efforts to combat the rate of improvement that cyber-attackers have when attacking cyber-physical systems.

To develop adequate defenses, we must efficiently generate end-to-end models of systems and adversaries. Current simulators, such as NetSim [5] and Mininet [6], provide a partial view of the system, but fail to exhibit physical constraints and

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conditional operations across different components. On the other hand, network simulators like the NS3 simulator provide a detailed view on most layers of a network. These layers, including the application and transport layers, make up what we refer to as the network stack. Additionally, existing co-simulators provide little or no support for *perturb-and-observe* to model adversarial scenarios and generate benchmark datasets for downstream applications such as risk assessment, attack detection, and risk mitigation.

We present Network Attack Testbed in [Power] Grid (NATI[P]G[1]), a co-simulation environment for a distribution power grid network using state-of-the-art simulators. The code base is publicly available through github [7]. This co-simulator is used to generate attack scenarios that can enable researchers to understand how attackers could behave in a network given a set of goals. Our work builds upon past work where researchers modeled attacks using network simulators to understand the behavior of attackers [8], [9]. Our framework also provides the ability to quickly tune attack scenarios through configuration files. This enables more users with little to no experience working with NS3 to use our framework straight out of the box. Finally, by modeling potential adversary behaviors, researchers can develop faster ways to identify them on the network during attacks.

Our primary focus is on Man-In-The-Middle (MIM) attacks on a power grid. We implement these attacks using the NS3 network simulator and evaluate their impact using the GridLAB-D simulator. Our scenarios extend beyond the transport and session layers of the Open Systems Interconnection (OSI) model, demonstrating application layer perturbations in the communication layer. This aim is to provide an example of how NATIG can be utilized by researchers without requiring specialized hardware.

Utilizing our testbed, we have identified distinct behaviors between grid-following and grid-forming inverters during set-point attacks initiated by a MIM attacker. These findings have been validated in previous work by Du et al. [10] in the context of islanding attacks. Additionally, our testbed has been used to determine capacitor and generator settings that minimize frequency deviation when switches are tripped and microgrids are islanded.

The contributions of the paper are as follows:

- With the use of NS3 as an intermediate network, our co-simulation tackles the entire network stack, creating a simulation close to what is expected of a real test bed. NS3 provides detailed simulations of real-world network protocols and scenarios with greater flexibility and a rich set of libraries, facilitating comprehensive network analysis and research.
- We produce a containerized framework for a wide range of cyber resilience assessment applications, improving upon existing testbeds.
- We demonstrate the feasibility of modeling, simulating, and validating CPS environments without using specialized hardware.
- We leverage application layer commands using DNP3 protocol in NS3 to simulate realistic grid behaviour.
- We demonstrate impact of an attack introduced in the network layer on the physical layer through MIM attack.

This paper is organized as follows. Section II describes the motivation behind implementing a security simulation platform. In Section III, we survey existing approaches and address our motivation. Section IV delineates the design of our framework and use cases of cyberattacks. Section V describes the details of attack scenarios to demonstrate the usability of our framework. We report our experimental results in Section VI. Section VII concludes our work and discusses future work.

II. MOTIVATION

The integration of GridLAB-D and NS3 allows users to simulate and analyze the interaction between physical grid components and communication networks, providing a comprehensive view of grid operations. This multi-layer architecture permits the detailed study of control strategies and their impact on grid stability and performance under various scenarios, including cyberattacks. Additionally, users can experiment with different network topologies, communication protocols, and security measures, making the platform a versatile tool for both academic research and practical utility management.

By enabling NATIG to work with both the IEEE 123 feeder model and the IEEE 9500 bus model with minimal to no changes needed by the user in the code, NATIG allows for a more extensive and scalable NS3 setup, showcasing the framework's flexibility to handle larger, more complex networks compared to the IEEE 123 bus model. Users benefit from the ability to tailor the network division to their specific needs, enabling detailed and customizable simulations. This flexibility enhances the platform's applicability for diverse research and operational scenarios, providing a comprehensive tool for grid management and analysis.

NATIG has the ability to simulate complex cybersecurity attacks, such as Man-in-the-Middle (MIM) and Distributed Denial-of-Service (DDoS) attacks, within a DNP3 application on NS3 with needing much setup from the user. Users can simulate sophisticated cyberattacks on grid networks without needing extensive cybersecurity or network programming knowledge, thus lowering the barrier to entry. NATIG enables users to evaluate the resilience and security measures of their network under various attack scenarios, providing valuable insights into potential vulnerabilities.

In this paper we focus on MIM attacks. These attacks are easier to accomplish on a system with a wide surface of attack. The integration of distributed energy resources (DERs), such as solar panels and wind turbines, has expanded the attack surface of the grid. These devices often have less stringent security measures than traditional grid components,

[1] https://github.com/pnnl/NATIG/tree/master

making them more vulnerable to MIM attacks. This makes it challenging for grid operators to identify and respond to these attacks in a timely manner. Therefore, usage of our framework can allow system defense experts to simulate MIM attack scenarios to understand what to expect when monitoring their own systems.

III. BACKGROUND AND RELATED WORK

The power grid, a vital infrastructure influencing daily life, must balance supply with demand and manage unexpected events like extreme weather, which can disrupt its functionality and affect many people. Ensuring the grid's resilience [11] involves identifying and mitigating potential threats. Traditional power systems have transformed into cyber-physical systems [12], integrating with information and communication technology. As the use of network simulators grows, researchers increasingly rely on them to develop new technology. However, for effective evaluation and improvement, these simulators must evolve to accurately replicate real-world networks while maintaining simulation efficiency.

A. NS3

NS3 [13] is a network simulator commonly used in network architecture and system development. It can simulate almost all aspects of a network, including the physical properties of signal transmission. This allows users greater control over the different aspects of the network, allowing them to simulate various attack scenarios with different network topologies, interject data flows at a particular node, and replace normal data with false data. An example of this control is setting whether or not a packet has reached its final destination. This simulates how an attacker can stop a packet at the node controlled by them before sending out a new packet with different information to a victim node. Another example is updating a source Internet Protocol (IP) to the original source IP that was used by the intercepted packet, leaving a victim unaware of an IP change.

NATIG tool provides different network topologies (4G, 5G, mesh etc.) configurations out of the box which will be useful to end-users because they do not have to put in time and effort to build these topologies. Our system has a configuration file that can be used to set different topologies and tune attack parameters. The simulator can also collect performance and routing information using NS3 monitors. Using this tool, users can create attacker models that can be used as controllers in a network to adapt the network settings in response to attackers. NS3 has been used in the past to run realistic network scenarios [14], [15]. This means that any data collected using the NS3 tool as a network simulator can give indications to the user how their protocol or workload would behave on a real network. However, NS3 is still a simulator, which means that there is a need to understand how much noise a target network might have. Using NATIG, users can manipulate noise levels through configuration files, allowing them to tailor the simulation to reflect the potential for noise in a target network. Similarly, configuration files empower users to adjust the microgrid composition simulated using GridLAB-D.

B. GridLAB-D

GridLAB-D is a simulation tool that enables power distribution system simulation and analysis [16]. It is extensively used to represent the system behavior of different power system components and complex interactions between these components and modern grid technologies like DERs. It currently supports for the IEEE 123 and 9500 feeder models. The user can utilize either one based on their use case. With the recent interests in studies regarding the interaction of power system and communication networks, GridLAB-D has been used for bench-marking IEEE feeder models for modeling detailed and dynamic power system operations and time-series power flow simulations [17], [18]. It facilitates larger simulations of power system models and simplified implementation of system architectures for illustrating a wide range of scenarios that reflect the impacts of communication layers on power system operations. GridLAB-D is also used to show grid behavior under realistic conditions [19]. The seamless integration of GridLAB-D with NS3 (network simulator) through HELICS (co-simulation framework) forms the core of our grid workload co-simulation environment.

C. CO-SIMULATION DESIGN

There are existing platforms that simulate power systems. One of these platforms [20] includes a mechanism that uses NS3 as a networking interface in conjunction with Framework for Network Co-Simulation (FNCS) and GridLAB-D as tools to keep track of value changes over time of power system components. In line with this work, we utilize a co-simulation to simulate cyberattacks on power networks. Additionally, NATIG allows for larger simulation by using HELICS instead of FNCS as the interconnect between GridLAB-D and NS3 3. Battarai et al. [17] presents a HELICS based co-simulation environment, but the work does not focus on scenarios involving application layer perturbations. We introduce DNP3 protocol as part of NS3 to send measurement data and control commands between the NS3 nodes representing components in Supervisory Control and Data Acquisition (SCADA) systems.

Overall NATIG enables researchers and system administrators to understand the behavior of attackers on the grid. A user can use such a tool to gather information on the impact of an attacker on the behavior of nodes in the grid. Additionally, in many cases, conducting an attack on an actual network is not commonly allowed and many researchers are not able to access quarantined networks to run different attacks. Using a co-simulation environment such as our NATIG framework can be a be starting point that users use to understand the behavior of attackers. Finally, data containing the impact of attacks on the grid is scarce making the development of counter solutions near impossible. NATIG

can be used to provide more data to users on how attacks can evolve over time with higher granularity then data collected from a system. The generated data from NATIG can be instrumental in designing and implementing robust defense strategies for reconfigurable networks, which are becoming increasingly prevalent in the evolving power grid landscape.

D. RECONFIGURABLE NETWORKS

Live reconfiguration is essential for enhancing network security and performance in dynamic environments like 5G networks. Current networks can reconfigure themselves as workloads and needs change, such as user equipment, relay antennas or topology changes. In 5G networks, slicing allows network partitioning to isolate attacks, preventing them from affecting the entire network. These slices can be updated over time to isolate sections or distribute resources effectively. This makes live reconfiguration a powerful tool for security and performance improvements. A significant research area focuses on applying 5G network slicing and reconfigurability to power grids. NATIG provides user control over topology and node locations, enhancing simulation realism by considering node location to calculate interference in 4G and 5G networks. Users can input real network coordinates through a configuration file, which NATIG uses to place NS3 nodes on a 2D grid.

E. DNP3 PROTOCOL

DNP3 is a protocol used to send packets between nodes, commonly in a utility distribution network [21]. According to surveys, more than 75% of North American electric utilities use or have used DNP3 as a communication protocol [22]. Using this protocol in conjunction with the NS3 simulator allowed NATIG to develop an end-to-end power grid traffic simulator akin to realistic power grid setups. NATIG can simulate various performance and security scenarios affecting grid functionality, including node failures or network attacks. Using NATIG, users can also monitor the traffic flow between the nodes to observe any changes in the traffic between when the attack is happening and when the traffic is normal.

F. CYBERSECURITY SIMULATIONS PLATFORMS

Previous work has been done with the goal of generating cyber attacks, such as MIM attacks on LAN wireless network [23], WAN networks [24], hard connected networks [25], and VANET networks [26]. Additionally, researchers have looked at the effect of cyberattacks, like MIM and denial of service, on power systems, and have found that denial of service attacks have significant impacts on the run time of devices on the network [27]. Another example of attack generation platform is GridSTAGE [28], which can be used to simulate false data injection attacks. It provides a framework where the user can input the parameters of an attack and measure its effects on network traffic.

The drawback of existing approaches is that they need in depth knowledge of the tool to be able to use them efficiently. We focus on creating an easy to understand simulation tool that can be used by any researcher. The tool can be used to parameterize the attack that they want to simulate and obtain traffic and node performance information, including electrical characteristics of the node and packet loss on a path, from the microgrid components and the communication layer for characterizing the effect of the attack on the grid.

IV. NATIG CO-SIMULATION DESIGN

This section briefly describes the co-simulation platform developed for simulating the cyber-physical dynamics of the distribution grid and simulating attack scenarios at various parts in the grid. The platform combines various industry-grade and open-source tools such as GridLAB-D, NS3, and HELICS to emulate the different layers of the grid infrastructure. The platform uses the communication protocol, DNP3, for the grid communications. This paper focuses on generation of MIM attack behavior in distribution grids. We use a Docker container to distribute NATIG to other researchers. Docker was chosen since it allows us to create a shareable environment that can be easily installed by other researchers.

The container enables researchers that do not have access to a real power network to conduct research in an environment that mimics realistic grid scenarios. Using the *docker ps –size* command, we found that our docker container with all the code and data needed to run the attack scenarios has a size of 1.69 GB (virtual 14.4 GB). The first value is the amount of data (on disk) that is used for the writable layer of the container. The second value is the total amount of disk-space used for the read-only image data used by the container. Most modern computers have enough available memory.

A. NATIG CO-SIMULATION PLATFORM

Our current platform is divided into three layers: the control layer, the network or communication layer, and the physical layer. GridLAB-D is used for the physical layer, while NS3 is used to construct the connecting network. Finally the control layer is represented as the control node and it is controlled through the main simulation program. The control layer consists of one utility control center that receives measurements from and sends control commands to the respective microgrids in the physical layer. The network layer represents the communication medium that carries information between the control and the physical layer. This layer consists of various network topology and uses DNP3 as grid communication protocol between the control layer and physical layer.

The physical layer consists of the physical distribution grid using IEEE 123 bus test feeder model with various DERs. The IEEE 123 bus feeder model is further logically split into three microgrids, which can be configured in grid connected or islanded mode. NATIG also has the option to use a physical distribution grid using IEEE 9500 bus test feeder model with various DERs. This bus feeder mode is further logically split into 11 microgrids/substations.

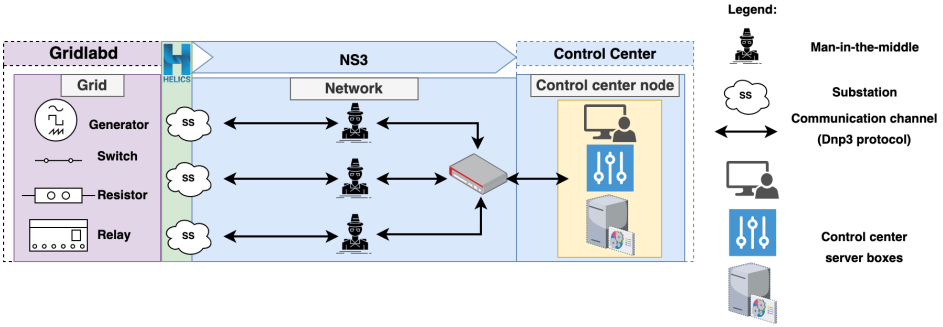


FIGURE 1. Overview of the co-simulation environment with interactions between HELICS, GridLAB-D, and NS3. Each node uses the DNP3 protocol to communicate. The control center, where the Open Platform Communications (OPC) server is located, is responsible for controlling a region of the grid network. The substation is responsible for power distribution and aggregating the information for the microgrid to be sent to the control center. The control center does periodic poll requests every 4 seconds. Once the poll request is received, the microgrid/substation responds with collected measurements.

TABLE 1. List of DER available in the IEEE9500 bus model.

DER Description			
DER name	location	Gen(ON/OFF)	Generation Setpoint
SteamGen1	MG6	ON	1000 kW
PVFarm	MG7	ON	500 kW
MicroTurb-1	MG9	ON	50 kW
MicroTurb-2	MG9	OFF	NA
MicroTurb-3	MG9	OFF	NA
MicroTurb-4	MG9	ON	100 kW
Diesel620	MG2	OFF	NA
Diesel590	MG4	OFF	NA
LNGEngine100	MG1	OFF	NA
LNGEngine1800	MG3	OFF	NA
Battery1	MG9	OFF	NA
Battery2	MG9	OFF	NA

Each microgrid has a substation that has remote terminal units (RTUs) that aggregate data and disseminate control signals coming from the control center to various DERs in the microgrid. The control center does periodic poll requests every 4 seconds. Once the poll request is received, the microgrid/substation responds with collected measurements. A MIM attack node sits between the control center and each of the substations. The MIM attacker changes the data that are sent between the substations and the control centers. The attacker's goal is to trick users into thinking the substations sent good data to the control center while sending a different command to the substation. The different command can be used to collect additional information from the substation or modify parameters such as tripping a relay in the microgrid.

1) GridLAB-D AND IEEE 123/9500 NODE TEST FEEDER

Our co-simulation tool utilizes two IEEE feeder models: a 123-node base model [18] and a larger 9500-node model [29] for scalability testing. Both models integrate Distributed Energy Resources (DERs) like solar PVs and diesel generators to create microgrids. These microgrids can be interconnected or operate independently, allowing researchers to simulate various grid configurations. Each

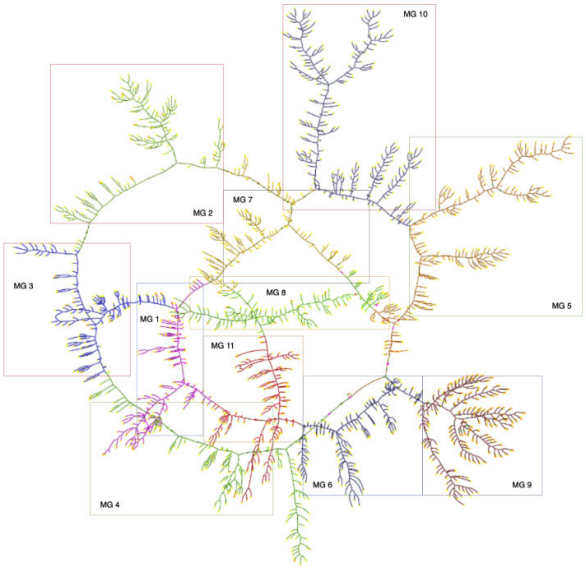


FIGURE 2. Overview of the IEEE 9500 bus model with microgrid divisions. This model has been divided in 11 microgrids, each of which are given an agglomeration point to gather data from the independent components of in the microgrid, like inverters or generators.

microgrid includes control systems and protective relays to manage power flow and ensure safe operation. The larger model incorporates 11 microgrids, as shown in Figure 2, with 15 DERs, enabling the study of complex power grid behavior under diverse conditions.

2) HELICS AND DNP3 WITH NS3

For our platform, as seen in Figure 1, we use HELICS as an interconnecting bridge between GridLAB-D and NS3. At the start of a simulation run, the HELICS-NS3 node and GridLAB-D connect to the HELICS broker as federates and are ready to send/receive data from/to each other. Each NS3 node is assigned a HELICS endpoint that is used to communicate with the HELICS broker. The HELICS broker

TABLE 2. List of the P and Q values for each individual microgrids in the IEEE9500 bus model.

Generation P and Q values		
MG #	Sum P value	Sum Q value
MG1	80430.0	0
MG2	80930.0	0
MG3	163820.0	0
MG4	85730.0	0
MG5	11420.0	0
MG6	0	0
MG7	563720.0	0
MG8	118280.0	0
MG9	1142580.0	0
MG10	84660.0	0
MG11	123090.0	0

serves as the timekeeper for the simulation. When the NS3 node receives a data request, it pings the GridLAB-D tool for the updated values for the registered points. Similarly, when the NS3 node receives a control command signal, it converts the command signal to GridLAB-D setpoint change request to the GridLAB-D tool via HELICS broker. Once the request is fulfilled, the broker advances the simulation to the next timestep. This continues until the end of the simulation.

We added DNP3 protocol into NS3 to simulate realistic grid communication scenarios between the utility control center and the distributed grid. The DNP3-NS3 module requires a configuration file consisting of all the measurement data expected from GridLAB-D simulation. The data points are GridLAB-D specific names of measurement data or set-points such as active power, voltage settings, ON/OFF states of switches etc., configured as analog or binary points. There are other types of points, but we focus on analog and binary points for the scope of this paper. To generate distribution grid scenarios, NS3 node with HELICS-NS3 and DNP3-NS3 modules enabled is needed. At the start of a simulation run, GridLAB-D measurement points get registered with GridLAB-D through the HELICS broker. HELICS-NS3 is responsible for retrieving measurements from the GridLAB-D tool and sending set-points to modify the DER settings. The DNP3-NS3 module is responsible for converting raw GridLAB-D measurement values into DNP3 protocol format and also translating DNP3 control commands into GridLAB-D specific set-point instructions.

B. NATIG CONFIGURATION

1) NETWORK TOPOLOGIES

NATIG takes a JSON file containing node to node connections, gives it to NS3, and uses it to build a topology. Normally, to make sure that the topology is valid, a user can use a topology generator such as an NS3 topology generator [30], but doing so requires prior knowledge of operating NS3. We simplify this process by reducing the knowledge needed to operate NS3 to a configuration file, thereby removing the need for additional programs outside the ones already installed in our Docker container.

Algorithm 1 Packet Interception Algorithm at the Stack. *ContToSub* Shows the Information Flow From Control Center to the Substation. *SubToCont* Shows the Information Flow From the Substation to the Control Center

victimAddr ← Address of the victim node
IsDestination ← *False*
function CapturePacket(p, addr)
  if *addr* = *victimAddr* **then**
    IsDestination ← *True*
  end if
end function
function ContToSub(AppHeader ActionID, UserData data, AppSeqNum seq)
  packet ← initiateReq(ActionID);
  transmit(packet, data, seq)
end function
function SubToCont(UserData data, *map* < *string*, *float* > *analog_points*, *map* < *string*, *uint_t* > *bin_points*, AppSeqNum seq)
  for k in *analog_points* **do**
    packet ← *k*
  end for
  for k in *bin_points* **do**
    packet ← *k*
  end for
  transmit(packet, data, seq)
end function
function SendNewPacket(DestID, SrcID)
  if IsDeatination **then**
    UserData *data.dest* ← *DestId*
    UserData *data.src* ← *SrcId*
    AddSeqNum seq ← current sequence number
    map < *string*, *float* > *analog_points* ← updated analog points
    map < *string*, *uint*16_*t* > *bin_points* ← updated binary points
  end if
end function

The configuration file allows control over the jitter of a node, the connection type between two nodes, and the topology of the network. For example, a user can create a configuration that generates a topology where there are two groups of nodes: The first group of nodes is connected using Carrier Sense Multiple Access (CSMA) following a mesh topology structure, and the second group is connected using a point to point connection using a star topology. Then both of the groups can be connected over Wi-Fi so that data can be sent between each cluster. This example is one of many configurations that can be created.

2) CYBER-ATTACK

We use JSON files to set up the attack in our setup. The configuration file takes in the start and end time of the attack, the number of attackers in the network, and attack specific

values. The attack-specific values include the parameters of the victim device that are being attacked, such as active or reactive power settings. It also includes the attack value that the attacker uses to update the value of the device parameter. Finally, the user can also select the attack scenario that they want to simulate on the network.

C. ATTACK GENERATION METHOD

To generate attacks, we use the internet module and the DNP3 module to intercept a packet and update it to contain new data. If the destination address is found to the be victim address, then the attacker simulates the packet arriving to its original destination. Then, through the DNP3 protocol, the attacker simulates sending a new packet containing the updated information or command. The *CapturePacket* function is set in the ipv4 l3 protocol module. The rest of the functions are developed through the DNP3 module. In Algorithm 1, the packet can be intercepted using the internet stack before it is sent to the target node(s).

1) OUR ATTACKER

In our experiments, the attacker is a Man-in-the-Middle (MITM) attacker who gains access to a network node using a DNP3 application within ns-3. The attacker utilizes a modified IPv4 Layer-3 protocol to determine if intercepted messages are destined for the victim's IPv4 address. Once identified, the attacker captures the messages and replaces the original data with updated information. Importantly, the source address is altered to match the original sender's address. Consequently, the receiver processes the message as though it originated from a legitimate source. This attack can operate independently to spread misinformation or be part of a larger strategy aiming both to spread misinformation and to take control of a node or section of the network.

In this attack scenario, the attacker intercepts traffic between the control center and the substation or microgrid. When intercepting packets from the control center to the substation, the attacker not only performs the requested action from the control center but also executes an additional malicious action of their choice. For instance, if the control center issues a poll request to network nodes, the attacker can intercept it, instruct the recipient to trip a relay, and modify the return data to conceal the changes. Conversely, by intercepting data from the substation to the control center, the attacker can send fabricated data to deceive the control center into believing that the substation is functioning correctly.

We are currently developing a new example allowing users to simulate a DDoS attack on the communication layer. In this attack, the attacker launches multiple applications that connect to a communication node, flooding the target path to degrade performance and increase data loss. This method can effectively isolate selected microgrids from the control center by having the network drop critical information preventing it from reaching the target node in a timely manner.

V. EXPERIMENTAL SETUP

This section describes generation of three types of attack scenarios on a distribution grid. We also demonstrate how to use NATIG to modify where the attacker is and how the attack impact varies depending on the location of the attacker in the network. For all of the three attack scenarios described in section V-C, the attacker is located between the control center and the substation.

A. DATA FORMAT AND SETUP

Understanding the experimental setup is crucial for assessing grid resilience and communication. The experiment includes a control center monitoring two scenarios: IEEE 123 test feeders with three microgrids, and IEEE 9500 test feeders with 11 microgrids. Grid dynamics are simulated using GridLAB-D. Microgrids can be islanded by tripping relays or switches. Each microgrid has a substation acting as an RTU, aggregating data and sending control signals to various DERs. The control center and substations, set up as NS3 nodes, communicate over a virtual NS3 network; substation nodes integrate with GridLAB-D via HELICS-NS3. Using the DNP3-NS3 module for DNP3-based communication, the control center issues periodic DNP3 polling requests every four seconds, with substations responding with data from GridLAB-D simulations. The control center can also send DNP3 control commands to adjust specific DER setpoints, which substations enact by relaying instructions to the GridLAB-D simulation.

During the configuration setup, all the GridLAB-D specific measurement data such as active power, reactive power, ON/OFF state of switches are preconfigured as DNP3 analog and binary points in a configuration file. During simulation start-up, these data points are ingested by the NS3-HELICS module at the substation node and registered with GridLAB-D. The raw measurements are collected and converted into DNP3 protocol format. The packaged data is then sent to the control center for processing over the network.

Figure 3 shows the system architecture for the IEEE 123 bus model with the design of our power grid, and the grid consists of the test feeder model divided in three microgrids. Figure 2 shows the system architecture for the IEEE 9500 bus model with the design of our power grid, and the grid consists of the test feeder model divided in 11 microgrids. The microgrids can be islanded by tripping the relay/switches connecting them to each other and the grid.The control center and the substation are also connected to the router where NS3 is loaded to allow different topologies during experimentation.

B. TOPOLOGIES TESTED

Evaluating different network topologies is crucial for improving grid resilience and security. We ran our attacks on three topologies: a default star topology, a ring topology, and a mesh topology configured via our topology configuration file. The star topology serves as the baseline, with a central

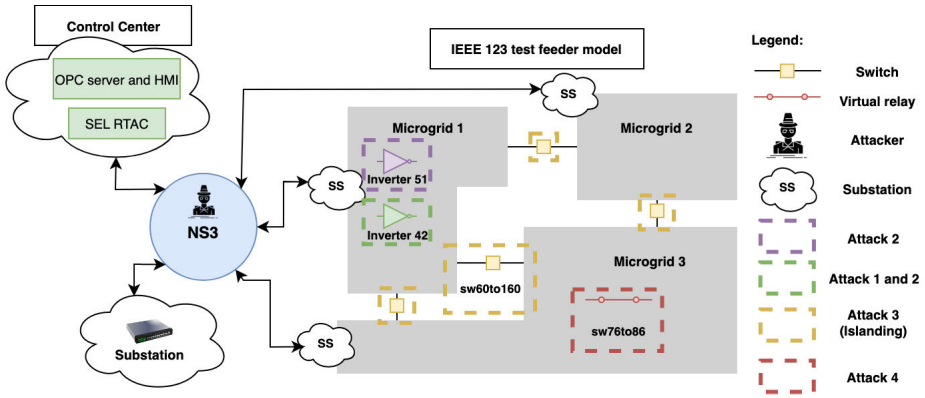


FIGURE 3. Overview of Microgrid setup for experimentation, using the IEEE feeder model as described by Ashok et al. [9]. We use this setup to run the cyberattacks and collect data on how the attacks impact the performance of the power grid. The attack conducts a MIM attack on two inverters in Microgrid 1, the switches connecting the microgrids, and the relay in Microgrid 3. Substations can be responsible for power distribution over multiple microgrids as well as act as aggregate points.

control center acting as a hub for all nodes; this setup is easy to manage but vulnerable to central node disruptions. In a ring topology, each node connects to two neighbors, offering redundancy since information can flow both ways around the ring if one link fails. Mesh networks provide the highest flexibility and resilience, with each node connecting to multiple others, thereby creating multiple data transmission pathways and making the grid highly resistant to individual node or link failures. The full detailed topology can be seen in [31].

C. ATTACK SCENARIOS

We place an attacker between the control center and the microgrid/substation. The attacker must stay unnoticed since it does not have authority to be on the network. Using a modified Internet Stack, the attacker reads the bytes in the packet and, if it finds a certain packet that matches the requirement for the attack, executes an interception. The attacker then modifies the intercepted packet data to communicate false information to the control center. This attack can be also used in conjunction with a command injection. Using both, an attacker can perform changes to the settings of a node while hiding it from the control center.

1) PACKET BYTE MODIFICATION

In this scenario, the attacker intercepts a poll request response going from the substation to the control center. The attacker then modifies the reactive power setpoints/reference (Qref) of Inverter 42 (grid-following inverter) situated in Microgrid 1 to trick the control center to believing that the inverter is in a different state than its current state.

2) ATTACK 1: INVERTER POWER SETPOINTS MODIFICATIONS

In this scenario, the microgrids are islanded from other microgrids and the grid. Then, a MIM attack changes the setpoint of an inverter to introduce voltage issues. The attack happens at approximately 2 minutes into simulation. As seen in Table 3, Inverter 42 has its Qref setpoint changed from 0 Var (default) to −50 kVar (attack value). In the second scenario, the microgrids are also islanded. Then, inverters are attacked consistently to cause voltage stability issues: Inverter 42 has its Pref value randomly toggled between 450 kW (default) and 350 kW (attack value) and Inverter 51 has its Pref value randomly toggled between 210 kW (default) and 110 kW (attack value). The attack starts 2 minutes into simulation.

This scenario uses packet capture (PCAP) files that are generated by NS3 to identify the attack and quantify its impact on the resulting network. *PCAP* files are used as ways to visualize network traffic. NS3 can generate them using the point-to-point helper class and point-to-point module and can be read using wireshark.

3) ATTACK 2: TRIPPING RELAYS USING COMMAND INJECTION

In this scenario, a command injection attack causes islanding of microgrids, as shown in Table 4. These relays are used in our experiment to island the microgrids from one another. The Virtual Relay is used as a load disconnect to cause additional UFLS. In some cases, power generation is enough to sustain the loads running on the microgrid, while in others (i.e., Microgrid 3) limited power generation increases the Under-Frequency-Load-Shedding (UFLS). UFLS occurs due to insufficient generation on the microgrids to supply the load. The sw60to160 relay was tripped via command injection at approximately 2 minutes into data capture. In another version of this attack, power dispatch has been adjusted to minimize UFLS. Finally, in a third version of this attack, the sw60to160 relay is tripped and the config file of the sw76to86 virtual relay was modified to trigger UFLS.

For this experiment, we use frequency measurement to compare the three parts. Frequency has been used to identify any load mismatches that may be happening in a power

TABLE 3. List of available Inverters in the IEEE 123 bus model with default and attack values.

Inverter Description						
Inverter ID	location	type	Rated	Pref	Qref	attacked values
Inverter 51 (or 3)	MG1	grid forming	400 kW	210 kW	0 VAR	Pref (110 kW)
Inverter 42 (or 2)	MG1	grid following	600 kW	450 kW	0 VAR	Pref (350 kW), Qref (−50 kVAR)
Inverter 101(or 1)	MG2	grid following	180 kW	126 kW	0 VAR	NA
Inverter 105	MG2	grid forming	600 kW	300 kW	0 VAR	NA
Inverter 76	MG3	grid following	120 kW	84 kW	0 VAR	NA
Inverter 80	MG3	grid forming	100 kW	70 kW	0 VAR	NA

TABLE 4. List of relays connecting the microgrids to one another and to the grid in IEEE 123 bus model.

Relay/Breaker/Switch Description			
Relay/switch ID	location	default value	attacked value
sw60to160	grid-MG2	close	open
sw18to135	grid-MG1	close	open
sw97to197	MG2-MG3	close	open
sw54to94	grid-MG3	close	open
sw151to300	MG1-MG2	close	open
sw76to86	MG3	close	open

system [32], [33]. When a cyberattack occurs, its impact on the point value of nodes in a power system causes the frequency to shift away from its normal, pre-attack value.

4) ATTACK 3: MICROGRID ISLANDING (ATTACK 2) ON LARGER A MODEL

For this scenario, we will island a microgrid on the IEEE 9500 bus model. Because this model is still in development at PNNL, it does not have a predefined division in microgrids. This means that we had to start by dividing the nodes into microgrids. To do so we group the nodes by connections with requirements to have 11 microgrids. We decided on 11 microgrids since it allowed us to have more microgrids compared to the division in the IEEE 123 bus model, thus allowing us to get a larger NS3 setup between the microgrids, which serves to show how our framework can scale depending on the size of the network. However, we leave it up to the user of the framework to provide a correct division of the network into microgrids.

VI. RESULTS

Understanding how different attack scenarios impact microgrids is crucial for enhancing their resilience and security. During experimentation, we collected data to demonstrate NATIG's functionality and identified critical insights. Changing parameters like the active power of an inverter (Pref) revealed differences in voltage behavior between grid-following and grid-forming inverters. Adjusting a generator's nominal output power during an attack showed that microgrid frequency could return to normal. We conducted four key experiments: First, we showed how an attacker can exploit data packets to manipulate microgrid behavior. Second, we demonstrated how manipulating inverter settings can disrupt a microgrid, highlighting NATIG's capability to simulate subtle and multi-node attacks. Third, we simulated tripping relays to isolate microgrids, analyzing their resilience to frequency disruptions. Finally, an islanding attack on the larger IEEE 9500 model revealed a more severe frequency surge (160 Hz) compared to the 123-bus model (72.5 Hz), emphasizing the need for balanced microgrid design.

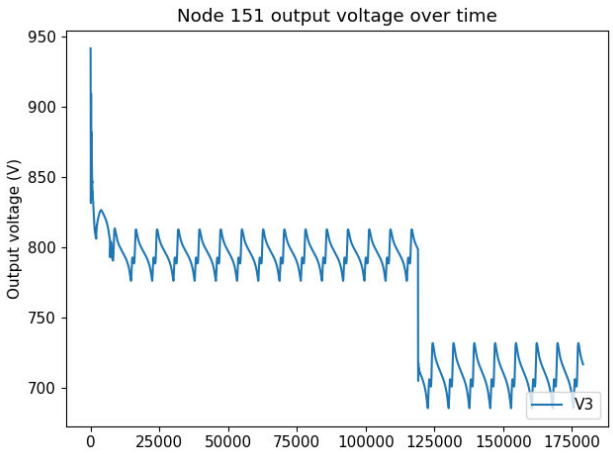


FIGURE 4. Overview of the voltage change once the attack starts at load 42, which is the load connected to inverter 42. The attack starts 2 minutes in the simulation.

A. PACKET BYTE MODIFICATION

Understanding substation responses is crucial for detecting vulnerabilities and improving grid security. We began the experiment by examining the three-segment, 274-byte response from the substation to the control center following a poll request, which includes node points and their values, indicating node behavior (e.g., generator output voltage) at the time of transmission. Each point is flagged to show its correctness. An attacker with access to these packets can analyze their history to map out the microgrid's nodes. In this experiment, we intercepted and altered several points within these response packets and then retransmitted them with new values. NATIG can generate PCAP files through the ns-3 simulator to visualize this attack, highlighting how exposed data can be manipulated to deceive network monitors.

B. INVERTER POWER SETPOINTS MODIFICATIONS

Figure 4 shows the impact of lowering the qref of a target inverter on Microgrid 1 on voltage C (V3) of a different node

(node 151) in the same microgrid. We saw similar impacts on the other voltages of the same node. When testing the attacks on a real system, we found that the real testbed uses the ARTEMIS solver as part of the HYPERSIM engine, on the other hand the gridlabd simulator uses the NR solver. We were not able to find a common solver between both setups. This causes some difference between the resulting voltage output since on one hand the ARTEMIS solver is EMT based rather than phasor based like the NR solver used by gridlab-d.

However, even with the data differing in terms of mean value, the data generated by NATIG does capture the trend visible when monitoring the performance of such a model using a real communication network. The average difference between before and after the attack on our real testbed is around 150 V whereas on NATIG it is around 100 V. With some tuning of the gridlabd model by some power grid experts, we can provide out of the box examples that can get pretty close to modeling the impact of attacks on actual hardware. Additionally, trained models, using datasets generated by NATIG, can identify counter attackers based on node-level impacts, as seen in the change of voltage measured at the phases of nodes. For instance, sudden inverter voltage changes could trigger traffic rerouting and attack notifications, isolating the compromised section and prompting countermeasures. As demonstrated in the second part of this attack (Figure 5), NATIG can also simulate multi-node attacks, where here, the attacker randomly varies the Pref values of two inverters on Microgrid 1.

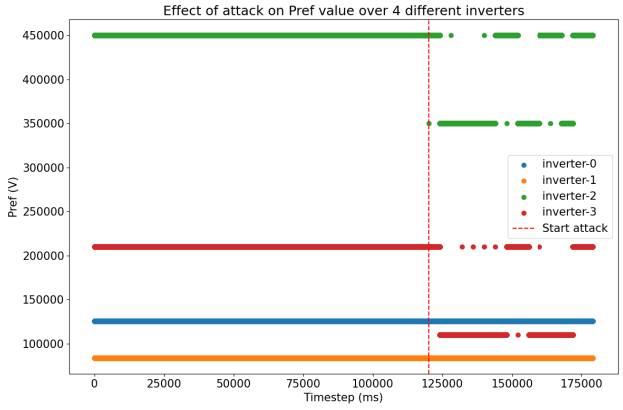


FIGURE 5. Overview of how the Pref value is randomly fluctuated for Inverter 42 (inverter-2) between 350 kW and 450 kW (the default value) and for Inverter 51 (inverter-3) between 210 kW and 110 kW after two minutes in the simulation.

Figure 6 shows a significant decrease in output voltage of Inverter 42 during the attack where the attacker randomly increases and decreases the Pref value. As seen in Figure 5, the variation in Pref value matches the variation in output voltage for inverter 42 (inverter-2). The output voltage drops to the attack value of 350 kW when the attacker sets the Pref value to 350 kW. By identifying these drop in voltages,

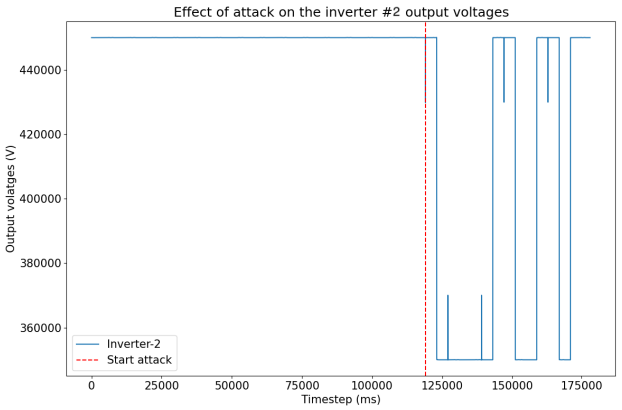


FIGURE 6. Overview of effect from attack on inverter 42's (inverter-2) output voltage. The attacker dynamically fluctuates the Pref value of two distinct inverters to affect the resulting output voltage of the inverter. In this scenario the microgrids are islanded from both each other and the grid.

a user of NATIG can create counter measures to such attacks by potentially resetting the inverter if the voltage fluctuates erratically. Finally, if we were to compare these results to the one where the Qref value was the set point that was attacked, we observe that the fluctuations happen more frequently and continue happening even after the attack. The difference in behaviour would help a user identify the type of setpoint that is being attacked.

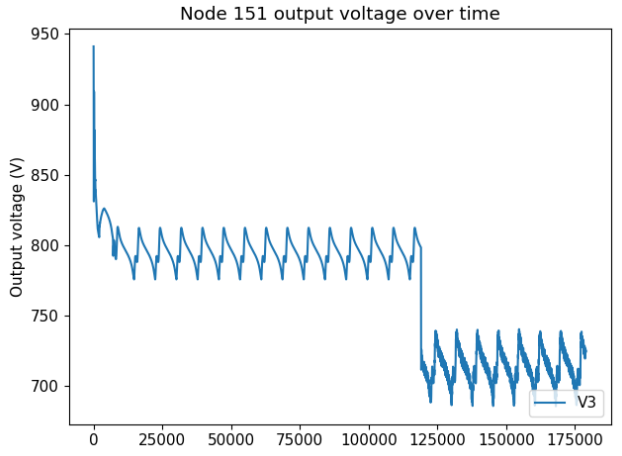


FIGURE 7. Overview of impact of attack on both Inverter 42 and Inverter 51 on an node 151's voltage. We can observe slightly more noise within the individual fluctuations compared to the data in Figure 4.

Figure 7 shows the affect of the attack on the output voltages of the load attached to Microgrid 1. Since Microgrid 1 is islanded from the rest of the microgrids and the grid when the attack starts, and the Pref value is randomly changed overtime, we can observe an increase in power fluctuation when looking at the output voltage of the inverters. This shows that both islanding the microgrid and changing the Pref values of the inverters will result in voltage impacts on other nodes of the microgrid.

TABLE 5. List of capacitors in Microgrid 3 in IEEE 123 bus model with the corresponding size at each of the phases A, B and C.

Microgrid 3 Capacitor Default values			
ID	phases	Nominal power	Capacitor size
cap83	A, B, C	2401.7771 V	A=200 kVAr, B=200 kVAr, C=200 kVAr
cap88	A	2401.7771 V	A=50kVAr
cap90	B	2401.7771 V	B=50kVAr
cap92	C	2401.7771 V	C=50kVAr

TABLE 6. List of available generators (synchronous and induction) available in the IEEE 123 bus model.

Synchronous Generator Default values			
ID	location	rated power output	delivered power
Gen1	MG1	10 MW	30 kW+3000 j
Gen2	MG1	1 MW	25 kW+8333 j
Gen3	MG3	450 kW	50 kW+16667 j
Gen4	MG2	600 kW	50 kW+16667 j

C. TRIPPING RELAYS USING COMMAND INJECTION

In this scenario, we trip the relays connecting the microgrids to one another, as seen in Table 4. We look at the frequency measurement of the microgrids to identify how resilient it is to attacks. Tripping the relays can cause the microgrids to become islanded from one another and the grid. In Figure 8, we can see that the frequency of the attacked microgrid increases to around 71 Hz during scenario A, the scenario where the generator and capacitor use the default value for NATIG, as seen in Table 6. Due to islanding, there is a deficiency of generation in Microgrid 3 and the frequency goes below the nominal value of 60 Hz. This leads to under frequency tripping (or shedding) of loads. Once the loads are shed, there is excessive generation in the microgrid as the dispatch is not adjusted in this case (dispatch being the outputs of the generators and the capacitor setting in this microgrid) which in turn leads to an overfrequency situation above 60 Hz.

Capacitors are a useful tool when regulating the frequency of a microgrid during an attack. Table 5 indicates the capacitor banks located in Microgrid 3. When the simulation ran with a larger capacitor and a generator with a lower nominal power rating and lower power amount delivered to interconnected nodes, as seen in Table 7, the frequency returned to the normal pre-attack value after approximately 2 minutes. This is almost the same as the previous scenario except that the generator dispatches are adjusted this time to match the load after the UFLS, which makes the frequency to go back to the nominal value of 60 Hz.

Finally, when we trip the virtual relays between nodes 76 and 86, as seen in Figure 3, while keeping the changes created in scenario B, we can see that the frequency does not return to the value before the attack started. This has additional underfrequency load tripping on top of what already happened in the first scenario in this attack leading to excessive generation in the microgrid and subsequently, the frequency going above 60 Hz. We also observe that after

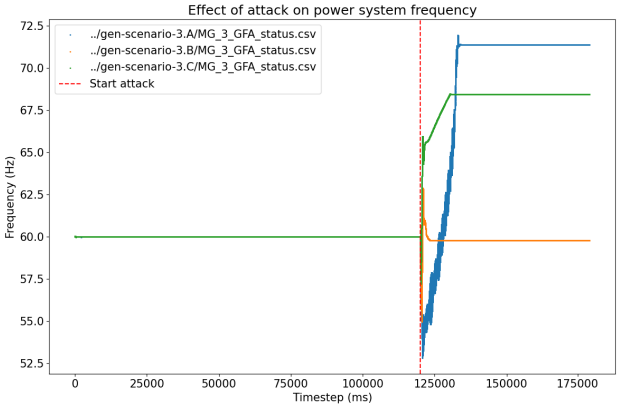


FIGURE 8. Overview of frequency changes depending on when the attack starts and what changes, if any, are done to reduce the frequency deviation. Scenario A (Blue) has no counter measures implemented. Scenario B (Orange) has initial islanding attack happening with the countermeasures implemented. Finally, Scenario C (Green) is similar to Scenario B with additional attacks happening to cause additional load shedding.

TABLE 7. List of phases of targeted capacitor that are updated during Microgrid Islanding attack example.

Updated Generator and Capacitor values			
ID	location	rated/nominal power	delivered power/ capacitor size
cap83	A, B, C	2401.7771 V	A=600 kVAr, B=600 kVAr, C=600 kVAr
Gen3	MG3	300 kW	20 kW+16667 j

an initial climb, the frequency slightly drops before starting to climb at a slower rate. The frequency finally stabilizes at around 140 seconds.

D. MICROGRID ISLANDING ON THE 9500 BUS MODEL

Figure 9 showcases an instance of an islanding attack on a power system model, specifically the IEEE9500 bus model. In this scenario, the microgrids are formed by grouping together a minimum number of nodes, without considering ensuring a balanced distribution of generation and load within each microgrid. This imbalanced setup leads to a significant rise in the network's frequency, reaching 160 Hz, compared to the standard 72.5 Hz observed in Microgrid 3 of the IEEE123 bus model.

This experiment highlights the potential issue of imbalanced microgrid configurations. By using this framework and observing the resulting frequency behavior, users can gain valuable insights into potential imbalances within their microgrid setup. This knowledge can then be used to guide them in selecting the most appropriate tuning approach when designing the division of their power system model. In essence, the experiment emphasizes the importance of considering both the number of nodes and the balance between generation and load capacity when designing microgrids. An imbalanced setup can lead to unintended

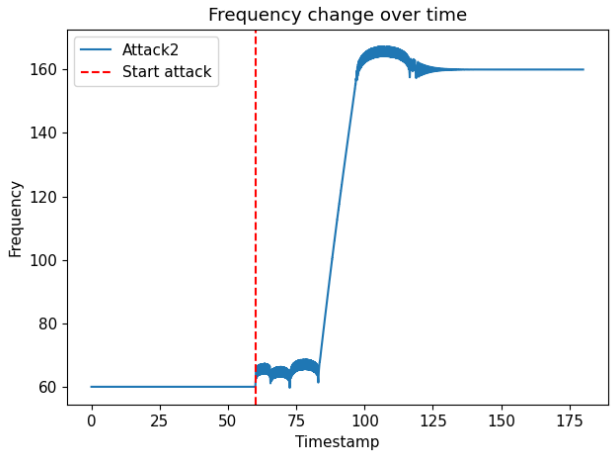


FIGURE 9. Overview of the frequency change when microgrid 9 is islanded by the attacker. Since there is a significantly larger generation than load, the graph shows a significant increase in the frequency resulting in a frequency of 160 Hz.

consequences, such as significant frequency deviations, and this experiment provides a way to identify and address such potential issues.

VII. CHALLENGES AND LIMITATIONS

Currently, NATIG operates entirely within the simulation space, but we plan to demonstrate its effectiveness in simulating attacks on hardware-in-the-loop systems. To achieve this, we are working on integrating the TapBridge module from NS3, making it configurable through user-modifiable configuration files. This module will allow users to pass data between physical devices through the NS3 network. Keeping NS3 in the loop facilitates rapid prototyping of network topologies and attack scenarios without requiring specialized hardware.

Although DNP3 is widely used in power systems, it's not the only protocol in use. Many industrial systems employ protocols like IEC 61850, Modbus, and proprietary protocols. Modeling only DNP3 can limit the ability to simulate environments using these other protocols. Additionally, different communication protocols have distinct error recovery and retransmission strategies, and these need to be modeled accurately to understand their impact on grid stability and performance. Future developments aim to include diverse protocols, accurate timing models, detailed error handling, comprehensive cybersecurity simulations, and tight integration with physical grid models to enhance the realism and applicability of such simulation tools.

We successfully tested our attack scenarios on simulated networks, but users currently can't control some of the topology settings within the simulation tool for a 4G/5G network. We are working to make these settings adjustable through configuration files for more user control. Finally, the 9500 bus model being still in development means that there is no data executed on a real system for us to compare the output of our simulator too. As part of future work we will be using our hardware in the loop implementation of NATIG to examine the impact of attacks on the 9500 bus model to create a detailed fidelity study on the larger network. To that extent, we should also spend some time to get the generated mean voltage during attacks to get closer to the voltage measured on a real system.

VIII. CONCLUSION AND FUTURE WORK

We demonstrate the benefit of using a lightweight tool to model different security scenarios and their effect on grid nodes. We also described the design of NATIG, and how it uses the IEEE model from GridLAB-D to model grid components. Using NATIG, we identified that there is differing behavior between grid forming and grid following inverters when the Pref and Qref values are changed by an attacker. Grid following inverters are more affected by any changes done to the Pref value, while grid forming inverters can restabilize the output voltage of the inverter down to its original value. We also identified parameter settings that enable the microgrid to recover the original frequency value once the microgrid is islanded.

In the future, we will extend our power grid simulation tool designed to be accessible and user-friendly for researchers. A web-based interface and cloud computing platforms will allow researchers to run simulations without needing high-powered local hardware. We will add an integrated framework to guide users in identifying and implementing mitigation measures against cyberattacks, eliminating the need to learn complex underlying tools. Finally, we will add the ability to swap network simulators (Mininet/NS3), solvers (EMT/phasor) and integrate real hardware nodes using Hardware-in-the-Loop (HIL) capabilities to increase the tool's adaptability and allow researchers to validate control systems under realistic conditions.

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